

B01 Practical Mastery Check

Closed-book · non-gating · save the attempt before review

Protection contract

This check does not unlock or block Gate A. Use the existing B01 PostgreSQL lab/practice data, but keep the teaching book and protected review closed. Save your SQL, results, validation evidence and short defense before opening the review.

1. Grain contract

Choose one one-to-many relationship in the B01 lab. State both input grains, expected match pattern and the output grain. Predict the fan-out before running the join.

2. Safe measure

Produce one row per parent entity with a child-derived amount plus a parent-level amount. Prove the parent-level measure was not multiplied.

3. Staged transform

Use at least three named stages. Run one intermediate stage independently and save its row/key control.

4. Deterministic window

Rank or select rows within a partition with a complete tie-break. State what would be nondeterministic without the final ordering key.

5. Time boundary

Use a fresh half-open [start,end) timestamp interval. Save count/min/max and test one boundary value.

6. Quality diagnosis

Write one orphan check and one duplicate-business-key check. Explain whether each is BLOCK or WARN and why.

7. Independent reconciliation

Create a control through a different path that could catch a row-loss, fan-out or financial error in task 2 or 3.

8. Execution + handoff

EXPLAIN one selective query; propose one candidate index without claiming speed. Finish with a concise handoff: grain, risk, validations, limitation and next owner.

Finished when

Your evidence contains the first attempt, query/result proof, at least two independent controls, one failure/recovery note and an own-words defense. Then open the protected mastery-check review, close it, and perform its changed retry.